Combinational Logic Tutorial

Total Questions: 20 — Time: 2 hours

Part A: Numerical Problems

Instructions: Solve the following problems. Show the Boolean simplification where necessary.

1. Implement the Boolean function $F(A, B, C, D) = \sum m(1, 3, 5, 7, 10, 11, 14, 15)$ using an **8:1 Multiplexer** with A, B, and C as selection lines. Determine the specific logic values $(0, 1, D, \overline{D})$ to be connected to data inputs D_0 through D_7.
2. Determine the minimum number of 2-input NAND gates required to implement a **Full Adder** circuit. Draw the gate-level diagram logic to justify your answer.
3. You are given two **3:8 Line Decoders** with active-low enable inputs. Explain the connections required to design a **4:16 Line Decoder**. Derive the logic required for the Most Significant Bit (MSB) to toggle between the two chips.
4. Design a combinational circuit that converts a 4-bit **Binary code** to a 4-bit **Gray code**. Write the simplified Boolean expression for each output bit (G_3, G_2, G_1, G_0).
5. In a 2-bit Magnitude Comparator comparing $A = (A_1A_0)$ and $B = (B_1B_0)$, write the simplified Sum of Products (SOP) expression for the output condition $A > B$.
6. Consider a **Ripple Carry Adder** designed to add two 4-bit numbers. If each Full Adder block has a propagation delay of 20 ns for the Sum bit and 10 ns for the Carry bit, calculate the total propagation delay to generate the final Sum bit (S_3) and the final Carry bit (C_{out}).
7. A 4-input Priority Encoder has inputs D_0, D_1, D_2, D_3 with D_3 having the highest priority. If the inputs are 1010 (where $D_3 = 1, D_2 = 0, D_1 = 1, D_0 = 0$), what is the binary output code?
8. Consider a 4:1 MUX where the select lines are $S_1 = A$ and $S_0 = B$. The inputs are fixed as follows: $I_0 = 0, I_1 = 1, I_2 = C, I_3 = \overline{C}$. Derive the simplified Boolean expression for the output Y in terms of A, B, and C.
9. Derive the minimal Boolean expression for the **Borrow** output of a Half Subtractor using only NOR gates.
10. A BCD adder adds two BCD digits. If the sum of the binary addition exceeds 9 (1001_2), a correction factor must be added. What is this binary correction factor, and what combinational logic gate arrangement detects the need for this correction?
11. Given the function $F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$, determine the number of **Essential Prime Implicants** (EPIs) using a Karnaugh Map.
12. Design a 3-bit **Odd Parity Generator**. If the input is XYZ, write the Boolean expression for the Parity bit P using XOR/XNOR notation.
13. Implement the function $Y = AB + CD$ using only **2-input NAND gates**. What is the minimum number of gates required?

14. Using a **1:4 Demultiplexer**, implement the Full Subtractor Difference output $D = A \oplus B \oplus C_{in}$. Explain which output lines of the DEMUX correspond to the minterms of the Difference function.

15. Given the generate function $G_i = A_i B_i$ and propagate function $P_i = A_i \oplus B_i$, write the expanded Boolean equation for the Carry output C_3 strictly in terms of input carry C_0 and the P and G terms.

Part B: Conceptual & Theory

Instructions: Answer the following based on theoretical principles and timing analysis.

16. Explain the concept of a **Static-1 Hazard** in a combinational circuit.

- Draw a K-map for a simple function that contains a static hazard.
- Demonstrate how adding a redundant prime implicant (consensus term) eliminates this hazard.

17. State **Shannon's Expansion Theorem**. Using this theorem, prove conceptually how any n-variable Boolean function can be implemented using a single $2^{n-1} : 1$ Multiplexer with one variable acting as a data input and the others as select lines.

18. Is the set of operators $\{\text{XOR}, \text{AND}\}$ functionally complete? Prove your answer by attempting to synthesize a NOT gate or a NOR gate using only these two operators.

19. Contrast the internal architecture of a **PLA (Programmable Logic Array)** versus a **PAL (Programmable Array Logic)**.

- Which of the two is generally faster and why?
- Which provides greater flexibility for complex Boolean functions with many shared product terms?

20. In a 64-bit adder, the propagation delay of a Ripple Carry Adder becomes prohibitive.

- Explain the theoretical limitation of the Ripple Carry architecture in terms of "gate depth."
- How does the Carry Look-Ahead architecture reduce this depth, and what is the hardware trade-off (cost) associated with this speed increase?